

Daily Logic Drill #4

Sheet by David Akinyele-Aje

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Disproof by Counterexample · Boundary Case Discovery · Derivative & Function Counterexamples · Universal Refutation.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Rapid counterexample discovery: minimal instances and boundary conditions.

1. Provide the minimal positive integer counterexample n to: “For all positive integers n , if n is prime, then $n^2 + 2$ is not prime.” (after TMUA 2022 Paper 2 Q3)
2. Find the smallest positive integer a for which there exist positive integers b, c such that $a \mid bc$ but $a \nmid b$ and $a \nmid c$. (after TMUA 2021 Paper 2 Q4)
3. Provide a function $f(x)$ with $f'(x) > 0$ for all $x \in \mathbb{R}$ that serves as a counterexample to: “If $f'(x) > 0$ for all x , then $f(x) > 0$ for all x .” (after TMUA 2018 Paper 2 Q6)
4. Find a real number $x > 0$ that serves as a counterexample to: “ $x > \frac{1}{x}$ for all $x > 0$.”
5. Provide a pair of real numbers (x, y) that serves as a counterexample to: “For all $x, y \in \mathbb{R}$, if $x^2 > y^2$, then $x > y$.”
6. Find the smallest positive integer n for which $2^n - 1$ is composite despite n being prime.
7. Provide a pair of non-negative real numbers (a, b) that serves as a counterexample to: “ $\sqrt{a+b} = \sqrt{a} + \sqrt{b}$.”
8. Find the smallest positive integer n that serves as a counterexample to: “ $n^2 + 1$ is always prime.”
9. Provide a differentiable function $f : [1, \infty) \rightarrow \mathbb{R}$ with $\lim_{x \rightarrow \infty} f'(x) = 0$ that disproves: “If $\lim_{x \rightarrow \infty} f'(x) = 0$, then $\lim_{x \rightarrow \infty} f(x)$ exists and is finite.”
10. Find the smallest odd composite integer n that is not a prime power.

Section B Manipulation Drills

(≤50 s each)

Structural counterexamples: functions, divisibility, and set operations.

1. A student claims: “If a polynomial $f(x)$ satisfies $f(n) \in \mathbb{Z}$ for all $n \in \mathbb{Z}$, then $f'(n) \in \mathbb{Z}$ for all

$n \in \mathbb{Z}$.” Provide a quadratic polynomial $f(x)$ that serves as a counterexample. (after TMUA 2017 Paper 2 Q16)

2. Give a pair of irrational numbers a, b whose sum is rational.
3. Give a pair of irrational numbers a, b whose product is rational.
4. Provide a pair of functions $f, g : [0, \infty) \rightarrow \mathbb{R}$ such that $f(x) \geq g(x)$ for all $x \geq 0$, but $f'(0) < g'(0)$. (after TMUA 2021 Paper 2 Q12)
5. Find the smallest positive integer n that is a counterexample to: “ $n^2 + n + 41$ is prime for all positive integers n .”
6. Provide a counterexample of three finite sets A, B, C showing that $A \cup B = A \cup C$ does not imply $B = C$.
7. Find the smallest positive integer n for which $3^n + 2$ is composite.
8. A student conjectures: “For every odd prime p , the integer $2^p - p$ is divisible by either 3 or 5.” Find the smallest odd prime p that serves as a counterexample. (after TMUA 2021 Paper 2 Q10)
9. Provide integers a, b and a modulus $m > 1$ showing that $a^2 \equiv b^2 \pmod{m}$ does not imply $a \equiv \pm b \pmod{m}$.
10. Give a counterexample function $f : [0, 1] \rightarrow [0, 1]$ showing that a continuous function need not have a UNIQUE fixed point.

Section C Substitution & Structure

(≤90 s each)

Authentic TMUA Paper 2 Prf1-counterexample multiple-choice questions.

1. Consider the statement: “(*) If a whole number n is 2 more or 4 more than a multiple of 5, then n is prime.” How many counterexamples to (*) are there in the range $1 \leq n \leq 50$? (after TMUA 2019 Paper 2 Q4)
2. Which of the following values of n is a counterexample to the claim: “For all positive integers n , if n is prime, then $n^3 - n$ is NOT divisible by 24?” (after TMUA 2022 Paper 2 Q3)
 - A) $n = 2$
 - B) $n = 3$
 - C) $n = 5$
 - D) $n = 7$
 - E) Both $n = 5$ and $n = 7$
3. A student claims: “If $f : \mathbb{R} \rightarrow \mathbb{R}$ is twice differentiable, $f(0) = 0$, and $f''(x) > 0$ for all $x \in \mathbb{R}$, then $f(x) \geq 0$ for all $x \geq 0$.” Which of the following functions serves as a counterexample? (after TMUA 2018 Paper 2 Q6)

- A) $f(x) = x^2$
B) $f(x) = e^x - 1$
C) $f(x) = x^2 - 3x$
D) $f(x) = \cosh x - 1$
4. Consider the statement: “For all positive integers a, b, c , if $a \mid bc$, then $a \mid b$ or $a \mid c$.” Which of the following triples (a, b, c) is a valid counterexample? (after TMUA 2021 Paper 2 Q4)
- A) $(2, 3, 4)$
B) $(6, 4, 9)$
C) $(5, 10, 15)$
D) $(7, 2, 14)$
5. Which of the following polynomials satisfies $f(n) \in \mathbb{Z}$ for all $n \in \mathbb{Z}$, but $f'(0) \notin \mathbb{Z}$? (after TMUA 2017 Paper 2 Q16)
- A) $f(x) = x^2 - x$
B) $f(x) = \frac{x^2+x}{2}$
C) $f(x) = \frac{x^3-x}{6}$
D) Both B and C
6. A student conjectures: “If a continuous function $f : [-\pi, \pi] \rightarrow \mathbb{R}$ satisfies $\int_{-\pi}^{\pi} f(x)dx = 0$, then $f(x) = 0$ for all $x \in [-\pi, \pi]$.” Which of the following is a valid counterexample? (after TMUA 2019 Paper 2 Q14)
- A) $f(x) = \cos x$
B) $f(x) = \sin x$
C) $f(x) = x$
D) All of A, B, C
7. Consider two statements about functions $f, g : [0, \infty) \rightarrow \mathbb{R}$ with $f(0) = g(0) = 0$:
- I. $f'(x) \geq g'(x)$ for all $x \geq 0 \implies f(x) \geq g(x)$ for all $x \geq 0$.
II. $f(x) \geq g(x)$ for all $x \geq 0 \implies f'(x) \geq g'(x)$ for all $x \geq 0$.
- Which of I and II are true? (after TMUA 2021 Paper 2 Q12)
- A) Neither
B) I only
C) II only
D) Both I and II
8. Consider the theorem: “An odd prime p can be written as the sum of two integer squares if and only if $p \equiv 1 \pmod{4}$.” Which of the following would constitute a counterexample to the **converse** of this theorem? (after TMUA 2018 Paper 2 Q17)

- A) A prime $p \equiv 1 \pmod{4}$ that cannot be written as the sum of two squares.
- B) A prime $p \equiv 3 \pmod{4}$ that can be written as the sum of two squares.
- C) An integer $n \equiv 1 \pmod{4}$ that cannot be written as the sum of two squares.
- D) A prime $p \equiv 1 \pmod{4}$ that can be written as the sum of two squares.

Section D Challenge

(TMUA / SMC difficulty)

Insight over computation. Each question has a short, elegant solution — find it.

1. Consider: I. “ $n = 41$ is a counterexample to ‘ $n^2 + n + 41$ is always prime’.” II. “ $n = 40$ is the smallest counterexample to ‘ $n^2 + n + 41$ is always prime’.” III. “ $n = 4$ is a counterexample to ‘ $n^2 + n + 41$ is always prime’.” Which of I, II, III are true?
 - A) none
 - B) I only
 - C) II only
 - D) III only
 - E) I and II only
 - F) I and III only
 - G) II and III only
 - H) I, II and III
2. Prove by contradiction: “There do not exist positive integers a, b with $a^2 - b^2 = 1$.”
3. A student claims: “For every positive integer n , at least one of $n, n + 2, n + 4$ is divisible by 3.” Determine, with proof, whether this claim is true.
4. Give a counterexample to: “If p is prime, then $2^p + 1$ is prime.”
5. Statement (*): “For every positive integer $n \geq 2$, there is a prime strictly between n and $2n$.”
 - (a) Verify (*) for $n = 2, 3, 4, 5$ by direct computation. (b) Does this verification constitute a proof of (*)? Justify your answer.

“Speed comes from recognition, not from rushing. Every shortcut is a pattern you have internalised.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths